

Boredom begets boredom: An experience sampling study on the impact of teacher boredom on student boredom and motivation

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Background. Boredom is a common complaint among students. Boredom was previously found to be negatively associated with academic outcomes, such as academic motivation, strategies, and achievement. It is of interest to understand students' in-class boredom, especially factors that might exacerbate it.

Aims. The current study examines the influence of teacher's boredom on students' in-class boredom and learning experience. It aims to understand the relationship between teacher boredom, students' perceived teacher boredom, student boredom, and student learning motivation.

Sample. A total of 437 students (54.8% female, $M_{\text{Age}} = 14.5$ years, $SD = 1.6$) and 17 of their teachers (29.4% female, 76.5% 40 years old or below) participated in the study.

Methods. We conducted an experience sampling study, in which participants completed a 2-week diary. Data were analysed using multilevel modelling.

Results and Conclusions. Results from multilevel modelling of 2,675 post-class evaluations indicated that teacher boredom was negatively associated with students' motivation. However, the relationship between teacher boredom and students' perceived teacher boredom was not significant, suggesting that students did not accurately perceive whether their teacher was bored. Results from indirect effect analysis further revealed that students' perception of teacher boredom predicted student learning motivation through student boredom. In other words, perceiving teachers being bored promoted students' own feeling of boredom, which in turn reduced their learning motivation. Together, these results indicate that when a teacher is bored in class, or when students perceive that their teacher is bored, students would have lower learning motivation.

Opening the door to a classroom in the middle of a lesson, while the teacher is busy explaining one concept after another, you might find a scene of students looking bored, resting their heads on the desk, dozing off, doodling on their notebooks, or secretly playing with their phones. Boredom is an emotion commonly reported by students (e.g.,

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Goetz *et al.*, 2014; Mann & Robinson, 2009). Boredom impedes students' learning motivation and academic performance (e.g., Pekrun, Goetz, Daniels, Stupnisky, & Perry, 2010; Tze, Daniels, & Klassen, 2016). Whereas much attention has been given to students' experience of boredom in educational settings, little is known about teachers' boredom, especially how it might affect students' boredom and learning experience. Through experience sampling method, the current study investigated the interplay of teachers' boredom, students' perceived teacher boredom, students' boredom, and students' learning motivation.

Boredom in academic setting

State boredom is defined as an aversive state of wanting to, but being unable to, engage in a satisfying activity (Eastwood, Frischen, Fenske, & Smilek, 2012). In a review by Vogel-Walcutt, Fiorella, Carper, and Schatz (2012) on academic boredom, this emotion is characterized by its unpleasantness and low arousal. It occurs when people experience an objective state of low arousal, accompanied by a subjective experience of dissatisfaction, frustration, and disinterest. It is a distinct emotion (Van Tilburg & Igou, 2012, 2017) that fits the contemporary definition of emotions by possessing affective, cognitive, physiological, expressive, and motivational components (see Nett, Goetz, & Daniels, 2010).

The experience of boredom appears to be ubiquitous in educational settings. Daschmann, Goetz, and Stupnisky (2011) reported that around 44% of middle school students partly to strongly agreed that they frequently felt bored in mathematics class. A study also demonstrated that boredom can be more prevalent than anxiety among students (Goetz, Frenzel, Pekrun, & Hall, 2006). In the university setting, one study reported that more than half of the students found at least half of their lectures boring, and almost a third found most or all of their lectures boring (Mann & Robinson, 2009). The prevalence of boredom among students was further revealed by experience sampling data in samples of high school and university students (Goetz *et al.*, 2014; Larson & Richards, 1991). In addition to in-class experience of boredom (e.g., Daschmann, Goetz, & Stupnisky, 2014; Mann & Robinson, 2009), students can also feel bored when they are studying outside the classroom (e.g., Webster & Hadwin, 2015).

Student boredom, motivation, and academic performance

Given the prevalence of boredom in academic settings, researchers are interested in its effects on students' academic achievement. One of the markers of state boredom is a strong desire to change the current activity or escape from it (Martin, Sadlo, & Stew, 2006; Smith & Ellsworth, 1985; Van Tilburg & Igou, 2012). This motivational component of boredom may make it particularly detrimental in academic setting as learning, no matter inside or outside classroom, requires one's persisting attention and motivation. The control-value theory (Pekrun, 2006) posits that certain emotions can influence students' cognitive resources, learning motivation, the use of learning strategies, and thereby academic performance. Pekrun proposed a reciprocal relationship between achievement emotions and academic outcomes. According to this theory, boredom is a deactivating emotion which has deleterious effects on students' intrinsic and extrinsic motivation to learn, and academic achievement. It postulates that boredom drives students to avoid or even leave the academic setting. Boredom can thus reduce learning motivation, which in turn may impede students' academic achievement.

This theoretical framework is well supported by a series of empirical studies. Results from two longitudinal studies, with primary school (Putwain, Becker, Symes, & Pekrun, 2018) and college samples (Pekrun, Hall, Goetz, & Perry, 2014), converge to suggest a reciprocal negative effect between boredom and academic achievement. Using multiple waves of data from an entire school year, these studies demonstrated that higher level of academic boredom predicted poorer subsequent performance, and poor academic performance predicted subsequent boredom level.

Empirical evidence demonstrates that boredom erodes students' motivation and thereby harms students' academic achievement. Pekrun, Goetz, and Titz (2002) conducted a series of cross-sectional, longitudinal, and diary-based studies in secondary schools and universities, where they found a negative association between boredom and various motivation variables, including interest, intrinsic motivation, extrinsic motivation, academic effort as well as overall motivation to learn. This is corroborated by later studies. Pekrun *et al.* (2010) found a negative relationship between boredom, intrinsic motivation to learn, and academic performance scores. Likewise, in-class boredom was associated with lower lecture attendance and lower average grade points (Mann & Robinson, 2009). In Fritea and Fritea (2013), motivational regulation strategies had a moderating effect on the negative association between boredom and academic performance. More recently, in Tze *et al.*'s meta-analysis (2016), academic boredom has been shown to have a small-to-medium aversive effect on academic achievement as well as a moderate negative effect on academic motivation. Together, these demonstrate a robust relationship between students' boredom, learning motivation, and academic performance.

Teacher boredom as an antecedent of student boredom

Considering the detrimental effects of academic boredom, it is essential to understand its antecedents. The control-value theory suggests that two groups of appraisals, control and value appraisals, are antecedents of a student's emotions. These appraisals are made based on the learning environment, which includes teachers' behaviours (Pekrun, 2006). Daschmann *et al.* (2011) proposed seven situational precursors to classroom boredom, namely monotony, lack of meaning, opportunity costs, being over-challenged, being under-challenged, lack of involvement, and teacher dislike. As many of the precursors involve teachers' behaviours, teachers can play a central role in students' boredom experience in class. From an experience sampling study conducted by Goetz, Lüdtke, Nett, Keller, and Lipnevich (2013), teaching characteristics, such as presenting materials with enthusiasm and making sure that students pay attention, were negatively associated with student boredom in classroom. College students reported their lecturers' personal attributes, qualities, and teaching strategies as some of the factors determining whether a lecture is dull or interesting (Sharp, Hemmings, Kay, Murphy, & Elliott, 2017). Similarly, ninth-grade students described teachers' personality and habits of instructions as antecedents of their boredom (Daschmann *et al.*, 2014).

Apart from teacher's personal attributes and teaching styles, teacher's emotions can influence students' affective experience. Specifically, we postulate that teacher boredom could be a precursor to student boredom. Emotional crossover refers to the transmission of emotions from one individual to another in the same social environment (Härtel & Page, 2009). The first empirical indication for the emotional crossover between teachers and students came from a study by Bakker (2005) who examined how flow experience crossed over from music teachers to their students. The study found that teachers' flow had a direct positive influence on flow among

students. Likewise, Frenzel, Goetz, Lüdtke, Pekrun, and Sutton (2009), Frenzel, Becker-Kurz, Pekrun, Goetz, and Lüdtke (2018) conducted two separate longitudinal studies that examined the emotional transmission of teacher's enjoyment towards teaching and students' enjoyment in class. They found that students' perceived teacher enthusiasm mediated the positive relationship between teacher and student enjoyment. They argued that teachers who enjoyed teaching would communicate their enjoyment of a topic or learning task through observable enthusiastic teaching style. It was by observing their teachers' visible enthusiastic teaching behaviours that students acquire the enjoyment of learning that topic or completing that task. Their argument is congruent with the indirect crossover mechanism, which posits that emotional crossover is mediated by interpersonal exchange styles (Härtel & Page, 2009).

Recently, Chang and Cherng (2017) extended the aforementioned three studies to investigate the crossover of discrete positive (enjoyment, hope, pride) and negative (anger, anxiety, shame, hopelessness, boredom) academic emotions from teacher to students in a sample of 890 students from twelve junior high schools in Taiwan. Students rated their teachers' and their own emotions. Results showed that students' perceptions of teacher positive and negative emotions positively predicted their own positive and negative emotions, respectively. The interplay of teacher and student discrete emotions was possible due to unconscious emotional contagion, which is based on people's perception of social interaction partners' emotional states. At an unconscious level, a person automatically and unintentionally mimics another person's facial expressions and other non-verbal cues, and these imitations act as physiological feedback that leads the person to experience the corresponding emotions (Hatfield, Cacioppo, & Rapson, 1993). It follows that students' perception of their teacher, including their teachers' emotions, will influence their own emotional experiences in academic settings.

Most existing literature on academic boredom focuses primarily on students' experience, and little attention has been devoted to teachers' experience. Much about teachers' boredom experience remained unknown, let alone its impacts. Teacher boredom could be an antecedent of student boredom, considering teachers' self-report of feeling bored while teaching in class (Bishay, 1996; Hensley, Pekrun, Goetz, Frenzel, & Keller, 2014), as well as the theoretical and recent empirical evidence on emotional crossover. To the best of our knowledge, this is the first study examining teacher boredom and its impact on students' learning experience. Previous findings that boredom experienced by students in class has a greater deleterious effect on academic outcomes than that experienced while studying (Tze *et al.*, 2016), coupled with the importance of teachers on students' in-class learning experience (Gorham & Christophel, 1992; Skinner & Belmont, 1993), suggest a need for a closer examination on the impacts of teacher boredom.

Current Study

In view of the research gaps, current study examined the relationships between teacher boredom, students' perceived teacher boredom, student boredom, and student learning motivation in classroom context. The control-value theory and empirical findings on academic boredom both suggest an inverse relationship between boredom and motivation among students. Drawing upon these theoretical and empirical evidence, in addition to recent research on emotional crossover in classroom settings, we formulated three main hypotheses:

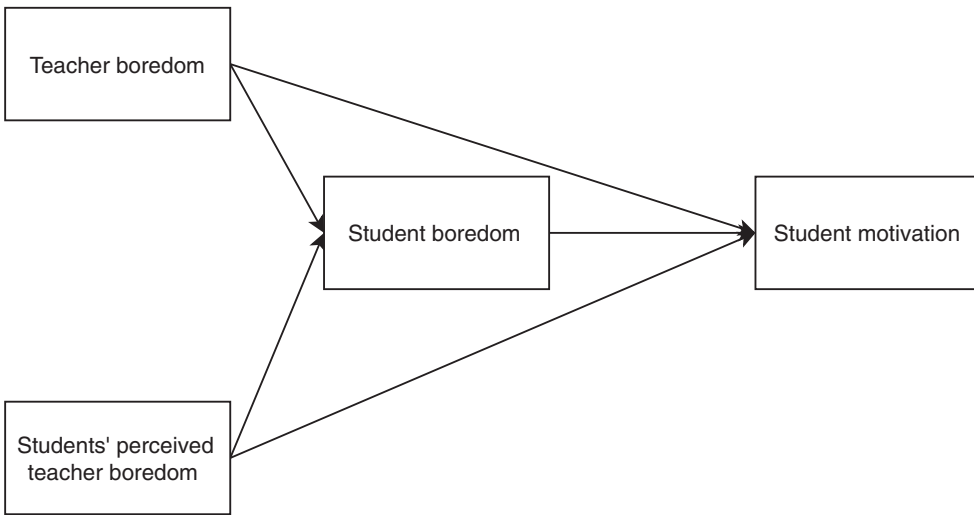


Figure 1. Hypothesized model.

- Hypothesis 1: students' perceived teacher boredom predicts student learning motivation through student boredom.
- Hypothesis 2: teachers' self-reported level of boredom predicts student learning motivation through student boredom.
- Hypothesis 3: perceived and actual teacher boredom exert indirect effects on student learning motivation through student boredom

Our hypotheses are depicted in Figure 1. We predicted that both perceived and actual teacher boredom are positively associated with student boredom and negatively with students' motivation to learn. To test our hypotheses, we used experience sampling method to collect data from a sample of students and their teachers across a 2-week period.

Method

Participants and design

Data from this study are derived from a larger research project on teachers' influence on their students. The study comprised of a convenience sample from two local schools in Hong Kong in November 2017. The schools were selected out of convenience sampling, and they are Hong Kong mainstream secondary schools. Only teachers who taught subjects that have classes nearly every day (i.e., Chinese, English, Mathematics, Liberal Studies) were invited to participate in the study; the length of each class ranged from 35 to 120 min. All students taught by the participating teachers were recruited for the study; those who did not give consent or opt out were excluded from the analysis ($n = 56$). A final sample of 17 teachers (29.4% female, 76.5% 40 years old or below) and 437 7th to 11th grade students (54.8% female, mean age = 14.47, $SD = 1.60$) participated in the study.

Procedure and materials

Data were collected over the course of 10 consecutive teaching days (i.e., 2 weeks) that did not overlap with the period of school examinations. Both teachers and students filled out baseline measures on the first day of the study and a brief questionnaire every day over 10 school days.

Teachers

All teachers completed their questionnaires online. Prior to the study period, participating teachers were first directed to fill out an online questionnaire about their demographics.

Throughout the 10 teaching days, the teachers followed a web link to complete a short questionnaire on Qualtrics. Teacher boredom was assessed by an item ('In general, during the core subject class today, I felt bored'), which was anchored on a 6-point Likert scale ranging from 1 (*strongly disagree*) to 6 (*strongly agree*). A notification of the web link was sent to each teacher at 12 noon every day through email, followed by a reminder at 8 pm.

Students

Paper-and-pen questionnaires were distributed to all participating students. Baseline questionnaires about the students' demographics were printed on perforated pages, bound in a booklet, and delivered to students participating in this study. The students were instructed to tear off, fill in, and submit the baseline questionnaire to the class representative on the first day of the study period.

Throughout the 10 teaching days, students were instructed to fill out a short questionnaire reporting their boredom level, perceived teacher boredom, and in-class learning motivation on the daily questionnaire immediately after each class taught by the participating teacher. We measured these constructs at the end of each class so as to reduce disruptions to daily teaching while capturing the influence of the participating teachers' emotions. Students were instructed to tear off the completed questionnaires and submit it to the class representative. The class representatives collected the questionnaires and put them into a sealed collection box placed in the general office every day.

In line with previous experience sampling studies in classroom settings (e.g., Becker, Goetz, Morger, & Ranellucci, 2014; Goetz *et al.*, 2013; Nett, Goetz, & Hall, 2011), we used single-item measures to avoid our study being overly intrusive to the classes. Student boredom during class was measured by an item 'In this class, I felt bored'. Students' perceived teacher boredom was assessed by parallel item wordings to those used for student boredom. Both measures were anchored on a 6-point Likert scale (1 = *strongly disagree*; 6 = *strongly agree*). Student learning motivation was assessed by an item adapted from the Task Value Scale in the Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich, Smith, Garcia, & McKeachie, 1991) ('I am very interested in learning the course materials'). A 6-point Likert scale ranging from 1 (*not at all*) to 6 (*very strongly*) was used for rating.

Ethics

All teacher and student consents were obtained prior to collecting their demographic data. Parental assent was also obtained for participating students. Participants were provided a debriefing sheet with information about the study after the data collection period. Ethical approval was received from The University of Hong Kong prior to data collection.

Statistical analysis

All statistical analyses were performed with *R*. Multilevel modelling (MLM) data analysis techniques were used to account for the nested structure of the data with 2,675 student daily ratings within 81 daily ratings of 17 teachers using the *lme4* package (Bates, Maechler, Bolker, & Walker, 2014) and *mediation* package (Tingley et al., 2017) in *R*. As the between-teacher variation is small ($ICC = .041$), a 2-level model accounting for 2,675 students' daily ratings nested within 81 teachers' daily ratings was conducted in the final analysis. Due to the constraints set by the multilevel modelling packages, missing data were handled with list-wise deletion. Mediation estimates were obtained through quasi-Bayesian approximation with 1,000 simulations.

Results

The descriptive statistics and correlations of the study variables are presented in Table 1. Teacher boredom negatively correlated with student learning motivation, $r(79) = -.27$, $p = .014$. However, the relationship between teacher boredom and students' perceived teacher boredom was not significant. Students' perceived teacher boredom significantly correlated with student boredom, $r(2658) = .50$, $p < .001$, and student learning motivation, $r(2618) = -.58$, $p < .001$.

Multilevel modelling

The ICCs in the unconditional model for teacher boredom (level 2) to students' (level 1) in-class boredom (Model 1) and student learning motivation (Model 2) were .100 and .107, respectively. MLM was used to test the association between teacher boredom and student boredom as well as student learning motivation. For within-day effects using MLM, we compared the fixed intercept (null) model with the random intercept model. The random intercept model for model 1 was not significantly better fitting than its null model, $\Delta\chi^2 = 2.924$, $p = .087$, and teacher boredom did not account for student boredom, $B = 0.080$, $p = .088$, 95% CI $[-0.012, 0.171]$. The random intercept model for Model 2 was significantly better fitting than its null model, $\Delta\chi^2 = 5.269$, $p = .021$, and teacher boredom was a significant predictor for student learning motivation, $B = -0.105$, $p = .023$, 95% CI $[-0.194, -0.016]$.

The ICCs in the unconditional model for students' perceived teacher boredom (level 1) to students' (level 1) in-class boredom (Model 3) and student learning motivation (Model 4) was .100 and .107, respectively. For within-day effects, the null model was compared

Table 1. Means, standard deviations, and correlations of study variables

	1	2	3	4
1. Teacher boredom				
2. SPT boredom	0.01			
3. Student boredom	0.14	0.50***		
4. Student motivation	-0.27*	-0.32***	-0.58***	
Mean	2.97	1.97	2.36	4.32
SD	0.83	1.12	1.27	1.15

Notes. SPT boredom = students' perceived teacher boredom.

* $p < .05$; *** $p < .001$.

with the random intercept model and subsequently with the random slope model. The random slope model was significantly better fitting for both Model 3 and Model 4 (Table 2). Students' perceived teacher boredom was a significant predictor to students' in-class boredom, $B = 0.557$, $p < .001$, 95% CI [0.503, 0.613]. It was also a significant predictor to student learning motivation, $B = -0.305$, $p < .001$, 95% CI [-0.357, -0.254].

Multilevel modelling mediation analysis was further used to examine whether student boredom mediated the relationship between the students' perceived teacher boredom and student learning motivation (Model 5). It was found the indirect effect was significant, $B = -0.271$, $p < .001$, 95% CI [-0.314, -0.232].

Discussion

In the present study, we found that students' perceived teacher boredom predicted student learning motivation through student boredom, which supports our Hypothesis 1. Students' perceived teacher boredom was positively associated with student boredom, whereas both students' perceived teacher boredom and student boredom were negatively associated with student learning motivation. Further, student boredom mediated the relationship between students' perceived teacher boredom and motivation (Hypothesis 3). For Hypothesis 2, we found a negative association between teachers' self-report level of boredom and students' motivation. Yet, surprisingly, teacher boredom neither related to student boredom nor students' perception of teacher boredom. Overall, these suggest that when teachers are bored, or when students perceive that teachers are bored, students would have a lower motivation to learn in class.

Out of our expectation, teacher boredom was not associated with student boredom. A possible explanation is that emotional crossover occurs only upon students' perceived teachers' emotions instead of self-report emotions experienced by the teachers. Prior research on emotional crossover in classroom settings examined students' reported perception of teachers' emotions rather than teachers' self-report emotions (Becker *et al.*, 2014; Chang & Cherg, 2017). Indeed, consistent with these studies on emotional crossover, we found a significant positive association between students' perceived teacher boredom and student boredom. In addition to that, we found an indirect effect of

Table 2. Chi-square differences tests between MLM models

	<i>df</i>	Deviance	AIC	BIC	$\Delta\chi^2$	Δdf	<i>p</i>
Model 3							
Fixed intercept (null)	3	8,508.8	8,514.8	8,532.4			
Random intercept model (RI)	4	7,859	7,867	7,890.5	649.76	1	<.001
Random intercept and random slope model (RIRS)	6	7,821.7	7,833.7	7,869	37.3	2	<.001
Model 4							
Fixed intercept (null)	3	7,994.1	8,000.1	8,017.8			
Random intercept model (RI)	4	7,785.3	7,793.3	7,816.8	208.832	1	<.001
Random intercept and random slope model (RIRS)	6	7,767.5	7,779.5	7,814.7	17.825	2	<.001

Note. AIC = Akaike information criterion; BIC = Bayesian information criterion; RI = random intercept; RIRS = random intercept and random slopes.

students' perceived teacher boredom on students' motivation through student boredom. In other words, perceiving teachers being bored may heighten students' levels of boredom and thus impede their motivation to learn in class. The positive association between students' boredom and motivation found in our study is consistent with past research (e.g., Pekrun *et al.*, 2010; Tze *et al.*, 2016).

The occurrence of emotional crossover upon perceived teacher emotions – and not teachers' self-report – may be attributed to the discrepancy between students' perception on teachers' emotional expression and teachers' actual experience. It is intriguing that students could not accurately perceive their teacher's emotion, as shown in our findings on the lack of association between teacher boredom and students' perceived teacher boredom. For emotional transmission to occur, emotions may have to be visually detectable (Becker *et al.*, 2014; Frenzel *et al.*, 2009, 2018). Teachers constantly regulating their emotions to facilitate teaching (Sutton, 2004) may make it hard for students to grasp their teachers' actual emotions.

Despite an absence of association between teacher boredom and student boredom, teacher boredom negatively predicted students' motivation in class. It is possible that boredom influenced teacher's teaching behaviours and hence students' learning motivation. Previous studies have provided evidence for the impact of teacher behaviours on students' motivation (Chang & Cherng, 2017). In a qualitative study where college students listed motivators and demotivators in learning, teacher behaviours accounted for around 44% of them. In fact, 'boring; teacher is bored with class' was one of the most frequently reported demotivators (Gorham & Christophel, 1992). In a longitudinal study (Skinner & Belmont, 1993), teacher behaviours, such as providing autonomy support and structure, were shown to predict three-to-fifth graders' motivation to engage across an academic year. In terms of the direct impact of teachers' emotion on students' learning, Patrick, Hisley, and Kempler (2000) found that teachers' enthusiasm served as a significant predictor of students' intrinsic learning motivation. The behavioural consequence of teacher's boredom may be manifold; they in turn may help explain variability in students' learning motivation. Future studies should further examine these links.

Limitations

Our results have to be considered with the shortcomings of the present study. First, we had a relatively small sample of teachers ($N = 17$) and the variability within teachers was small ($ICC = .041$), which undermines the reliability and generalizability of our findings. Our study may serve as a preliminary test of the relationships, although a larger sample size is suggested for future replication of the study's findings.

Second, we designed the questionnaires in a way to minimize disturbance to daily teaching. Whereas students completed the measures after each class, teachers reported their boredom level daily. As the students' questionnaires were in paper-and-pencil form, we could not record the time when students reported their emotions, neither could we control the time when teachers filled out the daily questionnaires. There might be a time lag between the experience in classes and their self-reports of emotions, which might harm the validity of our measures. Moreover, due to the practical need to keep the between-classes questionnaire brief, we did not ask our participants to fill out a comprehensive motivation scale. We adopted a single-item question on interest from the MSLQ because interest was found to be a major source of motivation in learning according to various theories and studies (e.g., Burke, 1995; Harackiewicz, Barron, Tauer, & Elliot, 2002; Harackiewicz, Tibbetts, Canning, & Hyde, 2014). Also, we only measured student

boredom at the end of each class but boredom level could differ at the start, middle, and end of a class. Future studies could consider employing personal digital assistant devices as in the studies by Goetz *et al.* (2014) and Nett *et al.* (2011) to record real-time data, and consider collaborating with school to resolve the time constraint and adopt a more comprehensive measure of motivation.

Third, regarding the association between students' perceived teacher boredom and student boredom, it is plausible that students might infer their teachers' feelings based on their own emotions. Since these two constructs were measured concurrently, it is possible that students might not have separated their own emotions from that of their teachers. As such, when they felt bored, they also perceived their teachers as bored. Although prior studies on emotional crossover also measured students' own emotions and their perceived teachers' emotions as two separate constructs (e.g., Becker *et al.*, 2014; Chang & Cherng, 2017), future studies are recommended to examine the extent to which students' developmental level of emotional awareness moderates the association between students' perceived teacher boredom and student boredom.

Fourth, we used a number of single-item measures to reduce disturbance to daily teaching. Although single-items have their obvious limitations, this procedure was frequently adopted in experience sampling studies in academic settings (e.g., Becker *et al.*, 2014; Goetz *et al.*, 2013; Nett *et al.*, 2011). Furthermore, our study is correlational in nature, the assumed causality between actual or perceived teacher boredom, student boredom, and motivation remain speculative. For example, it is possible that exposure to bored students leads to teachers' boredom. Future studies could adopt an experimental approach to elucidate their relationships.

Fifth, the length of each class varies according to subject, teacher, and day of the week; future studies are suggested to control for this variable. Sixth, given the limitations of the statistical packages that are currently available, our handling of missing data was not ideal. In the future, more sophisticated methods should be employed.

Implications

The current study demonstrated that, in classroom setting, students' perceived teacher boredom was related to students' experience of boredom even though students failed to accurately recognize teachers' feelings of boredom. Further, both actual and perceived teacher boredom undermined students' motivation. Several implications for research and educational practice could be drawn from these.

To the best of our knowledge, this is the first study investigating the influence of teacher boredom on students' learning experience. We only examined the effects of teacher boredom on students' levels of boredom and learning motivation in class. Future research could investigate its impact on other academic outcomes such as students' learning strategies and academic performance.

Moreover, our study contributes to the emotional crossover theory, that we tested both teacher self-report and student-perceived teacher boredom. From our results, although teacher boredom was not associated with student boredom, students felt more bored and reported lower learning motivation when they perceived their teachers being bored. Given the discrepancy between actual and perceived teacher boredom, our study brings up an interesting question for future research: What make students think that their teachers are bored? Past study proposed that collapsed body posture, moving the head backwards, and inexpensive movements as some physiological features of bored individuals (Wallbott, 1998), yet these seem to be inapplicable to teachers as they often

stand or walk around while teaching. If emotions like boredom (present study) and enthusiasm (Chang & Cherng, 2017) are transmitted upon students' perception of teacher emotions, future research should examine the factors affecting students' perception of teachers' emotions.

Last but not least, teachers are encouraged to cope with their feelings of boredom given the negative relationship between teacher boredom and students' learning motivation. A number of studies have been conducted on boredom coping among students (e.g., Daniels, Tze, & Goetz, 2015; Nett *et al.*, 2010, 2011) but not for teachers. According to a survey study with prospective teachers (Eren, 2016), cognitive-approach strategies moderated the negative association between boredom and engagement in classes. Teachers may benefit from adopting cognitive-approach strategies in regulating boredom. Future studies could further investigate the strategies for teachers to reduce their levels of boredom in class.

Conclusion

This experience sampling study has added potentially insightful findings to the realm of study of boredom from the perspective of how teachers' boredom plays a role in boredom in academic setting. Supplementing the literature on students' academic boredom and its possible associated impacts, with the inclusion of investigating teachers' boredom in this study, we provide support for the hypothesized relationships of actual and perceived teachers' boredom with students' boredom and their learning motivation. The results signified the importance of understanding the impact of teachers' boredom in the educational setting. Upon further research, we hope educators will be more aware of the importance of teachers' boredom and students' perception of teachers' boredom. Consequently, we may be able to thwart the toll that boredom takes among teachers and students.

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